

Luke Ferley

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Software Engineering student with an 83.4% academic average and hands-on experience developing backend applications, cloud-based solutions and data-driven software. Strong analytical and problem-solving skills developed through collaborative software projects involving databases, APIs, CI/CD, and testing. Experienced in Java, Python, SQL and AWS, with a strong interest in applying technology to solve real-world business problems.

EDUCATION

WeThinkCode

Occupational Certificate in Software Engineering – 83.4% Year 1 Average

Johannesburg, Gauteng

Sept. 2025 – Expected Dec 2026

EXPERIENCE

Peer Tutor Volunteer

WeThinkCode

Oct 2025 – Present

Johannesburg, Gauteng

- Selected based on top academic performance to provide technical mentorship and support to fellow software engineering students.
- Tutored 12+ students in Java, Python, OOP, and data structures; assisted in refactoring legacy (brownfields) codebases and architecting new (greenfields) systems.
- Facilitated programming workshops and technical mock assessments, guiding peers through debugging and structured problem-solving to reinforce code quality.
- Assisted students with Linux environment setup and configuration, as well as system troubleshooting and debugging.

Bootcamp Mentor Volunteer

WeThinkCode

Jul. 2026

Johannesburg, Gauteng

- Oversaw a group of five prospective students during a rigorous six-day selection bootcamp, tracking daily sprint progress, team engagement, and technical readiness.
- Mentored a team of six through Python fundamentals, collaborative version control, and algorithmic problem-solving to ensure successful group project delivery.

PROJECTS

Brownfields robot-world | *Java 21, EoDSQL, JDBC, SQLite, Docker*

Jun 2026 – Aug 2026

- Refactored and extended a concurrent TCP server-client codebase in a team of 5, introducing database persistence and a web API using Javalin, JDBC and SQLite.
- Improved overall code health from 9.24 to 9.5 (CodeScene) by systematically identifying and remediating code smells using the CodeScene CLI and web UI.
- Containerized the application with Docker and published images, and built an automated testing pipeline via GitLab CI/CD to support continuous integration.
- Iteratively developed a Makefile to standardize and simplify common development tasks throughout the 2-month project lifecycle.
- Practiced agile methodology through daily standups and weekly retrospectives across a 5-person team.

Spam Email Classifier | *Python, NumPy, Pandas, Matplotlib* | [GitHub](#)

Aug. 2026 – Present

- Engineered a spam email classification pipeline from a dataset of 190,000+ emails, performing exploratory data analysis, preprocessing, model training, and evaluation to achieve a best F1-score of 96%.
- Designed and executed a nested ablation experiment across four model versions, systematically isolating preprocessing changes to quantify their impact on classification performance.
- Implemented the classification algorithms, feature extraction, training, prediction, and evaluation pipeline from first principles without machine-learning libraries, using NumPy for numerical computation.
- Applied logarithmic-space probability calculations to prevent arithmetic underflow, improving numerical stability during model training and inference.

Serverless Notes API | *Python, AWS Lambda, DynamoDB* | [GitHub](#) | [OpenAPI Spec](#) | [Demo](#)

Jul. 2026 – Present

- Architected and deployed a serverless REST API for note management using AWS Lambda, API Gateway, and DynamoDB, achieving 95% code coverage through automated testing.

- Developed Lambda handlers with boto3 and built an automated unit-testing suite with pytest and moto to improve deployment confidence.
- Produced an OpenAPI 3.0 specification to support API adoption and third-party testing.
- Automated deployments via GitHub Actions using an IAM OIDC policy for keyless authentication to AWS.
- Implemented CloudWatch monitoring and API smoke testing with Postman and Insomnia to improve operational visibility.

Habit-Tracker API | *Java, Spring Boot, PostgreSQL* | [GitHub](#) | [API Docs](#)

May 2026 – Present

- Built a stateless REST API for habit tracking using Spring Boot and PostgreSQL.
- Secured user accounts with JWT authentication, Spring Security, and BCrypt password hashing.
- Developed a JDBC-based data access layer with parameterized SQL queries and Flyway-managed schema migrations, improving maintainability while reducing SQL injection risk.
- Validated application reliability through integration tests, Docker-based PostgreSQL environments, and API smoke testing with Insomnia.

TECHNICAL SKILLS

Languages: Java, Python, SQL, JavaScript, HTML, CSS

Cloud & Infrastructure: AWS Lambda, API Gateway, DynamoDB, IAM, CloudWatch, Docker, Linux/Unix, shell scripting

Data & Databases: PostgreSQL, SQLite, DynamoDB, JDBC, Pandas, NumPy, Matplotlib, Exploratory Data Analysis

APIs & Integration: REST APIs, OpenAPI, JSON, Postman, Insomnia

DevOps & Engineering: Git, GitHub Actions, CI/CD, Unit Testing, Integration Testing

CERTIFICATIONS & EXTRA LEARNING

AWS Educate: Getting Started with Storage, Introduction to Cloud 101, Introduction to Generative AI

CS50P: Introduction to Programming with Python, HarvardX

Microsoft: AI Fluency